Circuit Homework 1

Due day: Feb. 23

Question 1: Multiple-choice questions

(1) The voltage across a 1.1 kW toaster that produces a current of 10 A is:

(a) 11 kV (b) 1100 V (c) 110 V (d) 11 V

(2) The reciprocal of resistance is:

(a) voltage (b) current (c) conductance (d) coulombs

(3) An electric heater draws 10 A from a 120 V line.

The resistance of the heater is:

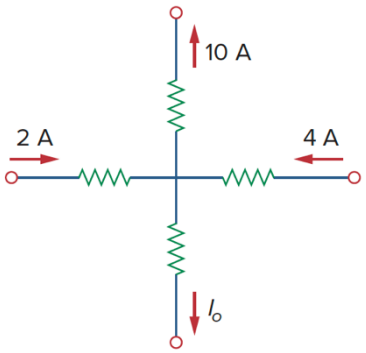
(a) 1200 Ω (b) 120 Ω (c) 12 Ω (d) 1.2 Ω

(4) The maximum current that a 2W, 80 kΩ resistor can safely conduct is:

(a) 160 kA (b) 40 kA (c) 5 mA (d) 25 μA

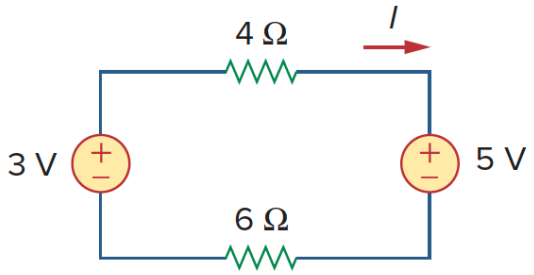
(5) The current I_o of

(a) −4 A (b) −2 A (c) 4 A (d) 16 A



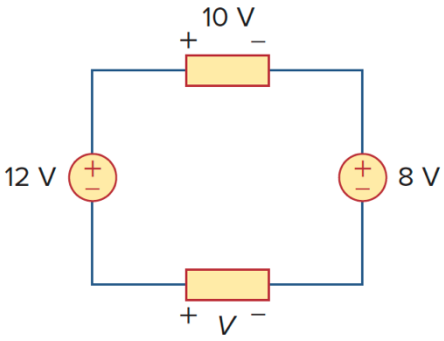
(6) The current I in the following circuit is:

(a) −0.8 A (b) −0.2 A (c) 0.2 A (d) 0.8 A

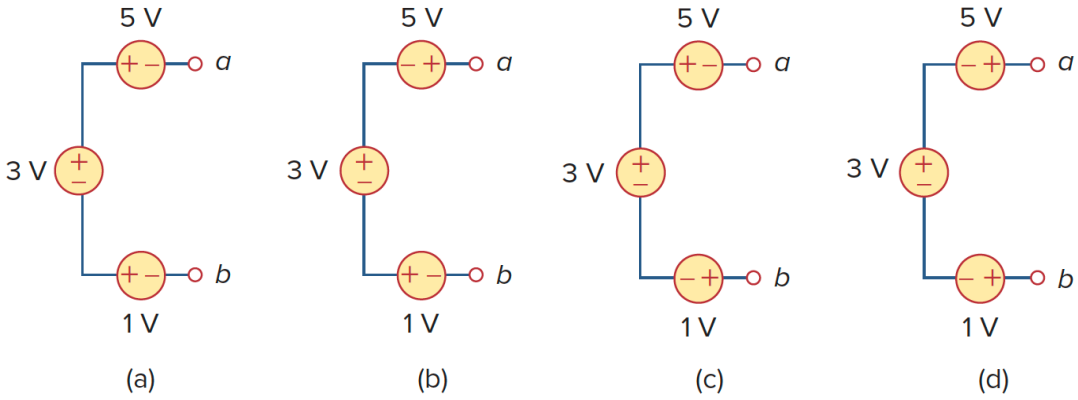


(7) In following circuit V is:

(a) 30 V (b) 14 V (c) 10 V (d) 6 V



(8) Which of the circuits will give you $V_{ab} = 7\,V$?

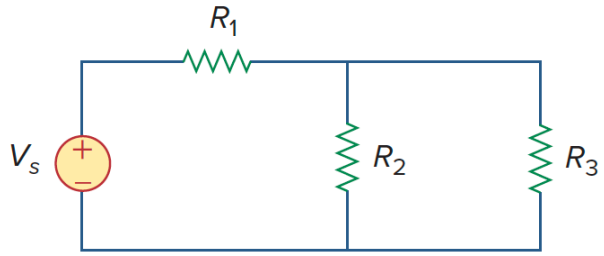


(9) In the following circuit, a decrease in R_3 leads to a decrease of,

select all that apply:

(a) current through R_3 (b) voltage across R_3 (c) voltage across R_1

(d) power dissipated in R_2 (e) none of the above



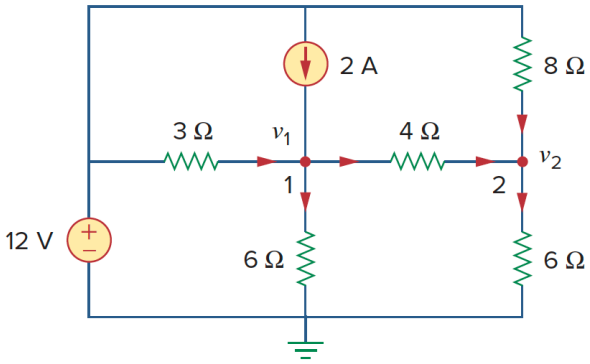
(10) At node 1 in the circuit, applying KCL gives:

(a) $2 + \dfrac{12 - v_1}{3} = \dfrac{v_1}{6} + \dfrac{v_1 - v_2}{4}$

(b) $2 + \dfrac{v_1 - 12}{3} = \dfrac{v_1}{6} + \dfrac{v_2 - v_1}{4}$

(c) $2 + \dfrac{12 - v_1}{3} = \dfrac{0 - v_1}{6} + \dfrac{v_1 - v_2}{4}$

(d) $2 + \dfrac{v_1 - 12}{3} = \dfrac{0 - v_1}{6} + \dfrac{v_2 - v_1}{4}$



(11) In the above circuit (in question 10), applying KCL at node 2 gives:

(a) $\dfrac{v_2 - v_1}{4} + \dfrac{v_2}{8} = \dfrac{v_2}{6}$

(b) $\dfrac{v_1 - v_2}{4} + \dfrac{v_2}{8} = \dfrac{v_2}{6}$

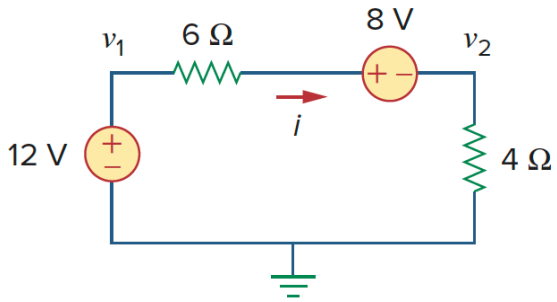
(c) $\dfrac{v_1 - v_2}{4} + \dfrac{12 - v_2}{8} = \dfrac{v_2}{6}$

(d) $\dfrac{v_2 - v_1}{4} + \dfrac{v_2 - 12}{8} = \dfrac{v_2}{6}$

(12) For the following circuit, v_1 and v_2 are related as:

(a) $v_1 = 6i + 8 + v_2$ (b) $v_1 = 6i - 8 + v_2$

(c) $v_1 = -6i + 8 + v_2$ (d) $v_1 = -6i - 8 + v_2$



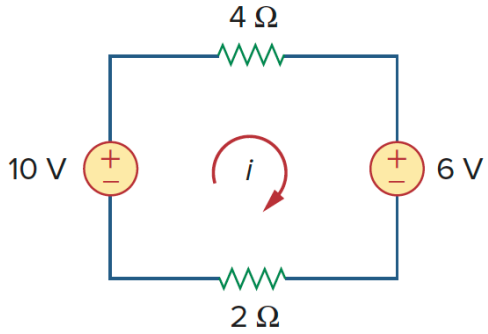
(13) In the circuit above (in question 12), the voltage v_2 is:

(a) −8 V (b) −1.6 V (c) 1.6 V (d) 8 V

(14) The loop equation for the circuit is:

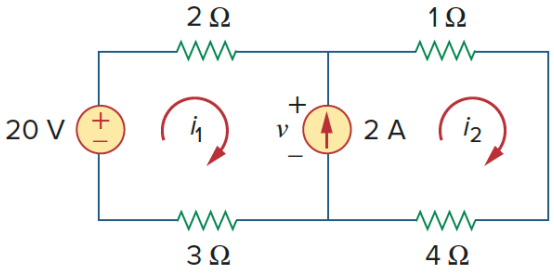
(a) $-10 + 4i + 6 + 2i = 0$ (b) $10 + 4i + 6 + 2i = 0$

(c) $10 + 4i - 6 + 2i = 0$ (d) $-10 + 4i - 6 + 2i = 0$



(15) In the following circuit, current i_1 is:

(a) 4 A (b) 3 A (c) 2 A (d) 1 A



(16) The voltage v across the current source in the above circuit (in question 15) is:

(a) 20 V (b) 15 V (c) 10 V (d) 5V

Question 2: Determine the current flowing through an element if the charge flow is given by:

(a) $q(t) = (4t^2 + 20t - 4)\ C$

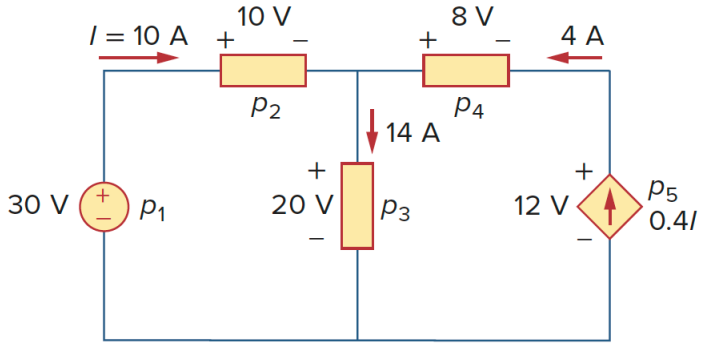
(b) $q(t) = (15e^{-3t} - 2e^{-18t})\ nC$

Question 3: Find the charge q(t) flowing through a device if the current is:

(a) $i(t) = 3\ A, \qquad q(0) = 1\ C$

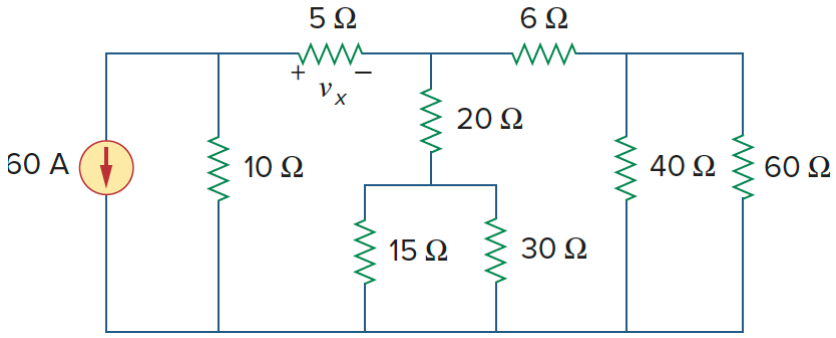
(b) $i\,(t) = (2t + 5)\ mA, \qquad q(0) = 0$

Question 4: Find the power absorbed by each of the elements in the circuit:

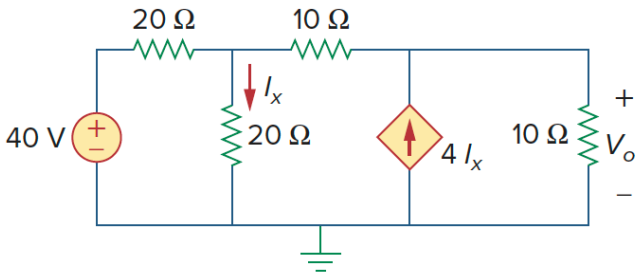


Question 5: A smartwatch uses a rechargeable battery rated at 3.7 V with a capacity of 200 mAh. A full charge allows the smartwatch to operate for 24 hours during active use, or 7 days in sleep mode. What is the average power consumption of the smartwatch in (a) active mode, and (b) sleep mode?

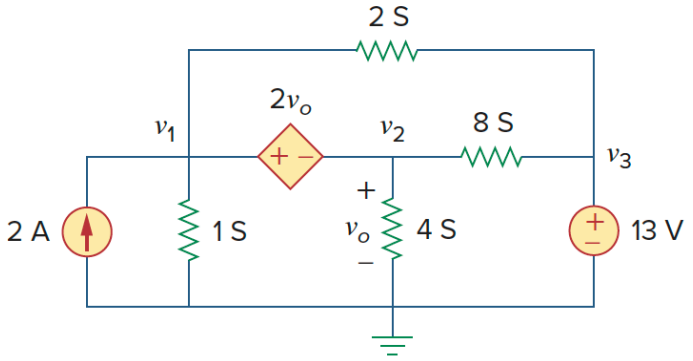
Question 6: In the following circuit, determine v_x and the power absorbed by the 60-Ω resistor.



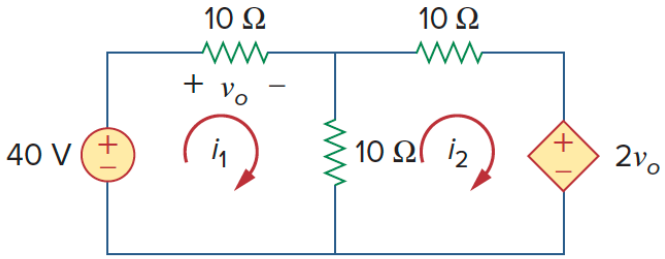
Question 7: Using nodal analysis, determine v_o in the following circuit.



Question 8: Determine voltages v_1 through v_3 in the circuit of using nodal analysis.



Question 9: Calculate the mesh currents i_1 and i_2 in the following circuit.



Question 10: Apply mesh analysis to the circuit and obtain I_0

